Pang-Wei Huang (黃邦維)

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Education

University of Illinois Urbana-Champaign — **Urbana, IL, USA**
Master of Computer Science (MCS) — *Aug. 2026 – Expected Jan. 2028*

National Taiwan University — **Taipei, Taiwan**
B.S. in Computer Science and Information Engineering — *Jun. 2025*

Technical Skills

Languages: C, C++, C#, Python
Technologies: Linux, Git, Docker, WPF/XAML, GitLab CI/CD, GitLab Runner, OpenMP, Pthreads

Work Experience

Alifecom Technology — **Taiwan**
Assistant Engineer, R&D Department — *Aug. 2025 – Jan. 2026; Jun. – Jul. 2026*

- Developed and maintained 3–4 WPF (C#/XAML) desktop applications for wireless equipment control and RF testing; adapted legacy GUIs to new hardware and refactored duplicated code to separate UI and device-control logic.
- Extended an existing test GUI with automated RF procedures specified by hardware engineers, integrating board communication via I^2C/Arduino/serial and Keysight MXA, MXG, and VNA instruments via SCPI over TCP.
- Enabled factory operators to perform previously engineer-operated RF tests at production scale, with an *estimated* 3–5$\times$ increase in testing throughput.
- Implemented an SCPI command parser in a customer-facing GUI for programmatic control of a wireless base-station simulator.
- Deployed an internal GitLab CE server on Ubuntu and configured a Windows GitLab Runner for WPF builds; implemented VM snapshot and scheduled GitLab data/configuration backups to NAS with retention policies.
- Evaluated and demonstrated AI coding agents (OpenCode, Claude Code, Codex), contributing to team adoption to support increasing customer demands amid hiring constraints.

Selected Projects

Can't Stop: Reinforcement Learning Agents | GitHub | Demo — *2026*

- Built a reproducible simulator and trained Ridge, Double DQN, and PPO agents to learn Stop/Continue decisions in the board game Can't Stop, holding dice pairing fixed to isolate the decision under study.
- Developed seeded, seat-swapped evaluations against traditional heuristics and prior agents, with game-level records and paired-bootstrap analysis.
- Exported lightweight policies for local JavaScript inference, enabling browser gameplay without a Python runtime or inference server.

Linux Kernel Programming — UIUC CS 423 — *Fall 2026*

- Built a Linux kernel module to monitor process CPU usage, using ProcFS, kernel timers, workqueues, linked lists, slab allocation, and synchronization primitives.
- Implemented a preemptive Rate-Monotonic Scheduler as a kernel module, including task registration/yield, admission control, a dispatcher kernel thread, wake-up timers, and priority-based preemption; completed and passed course tests for both projects.

Wireless Communications Experiment Reproduction | GitHub — *Jan. 2024*

- Implemented C++ simulations to reproduce published wireless communications experiments; used Python to analyze and visualize results, validating outcomes against theoretical predictions and independently generated randomized trials.